Folding from predicted contacts

Helico on MarinFold exp245's held-out FoldBench monomer sets

— 324 monomers scored by all 9 predictors · eval-val 95 / eval-test 210 / eval-denovo 19

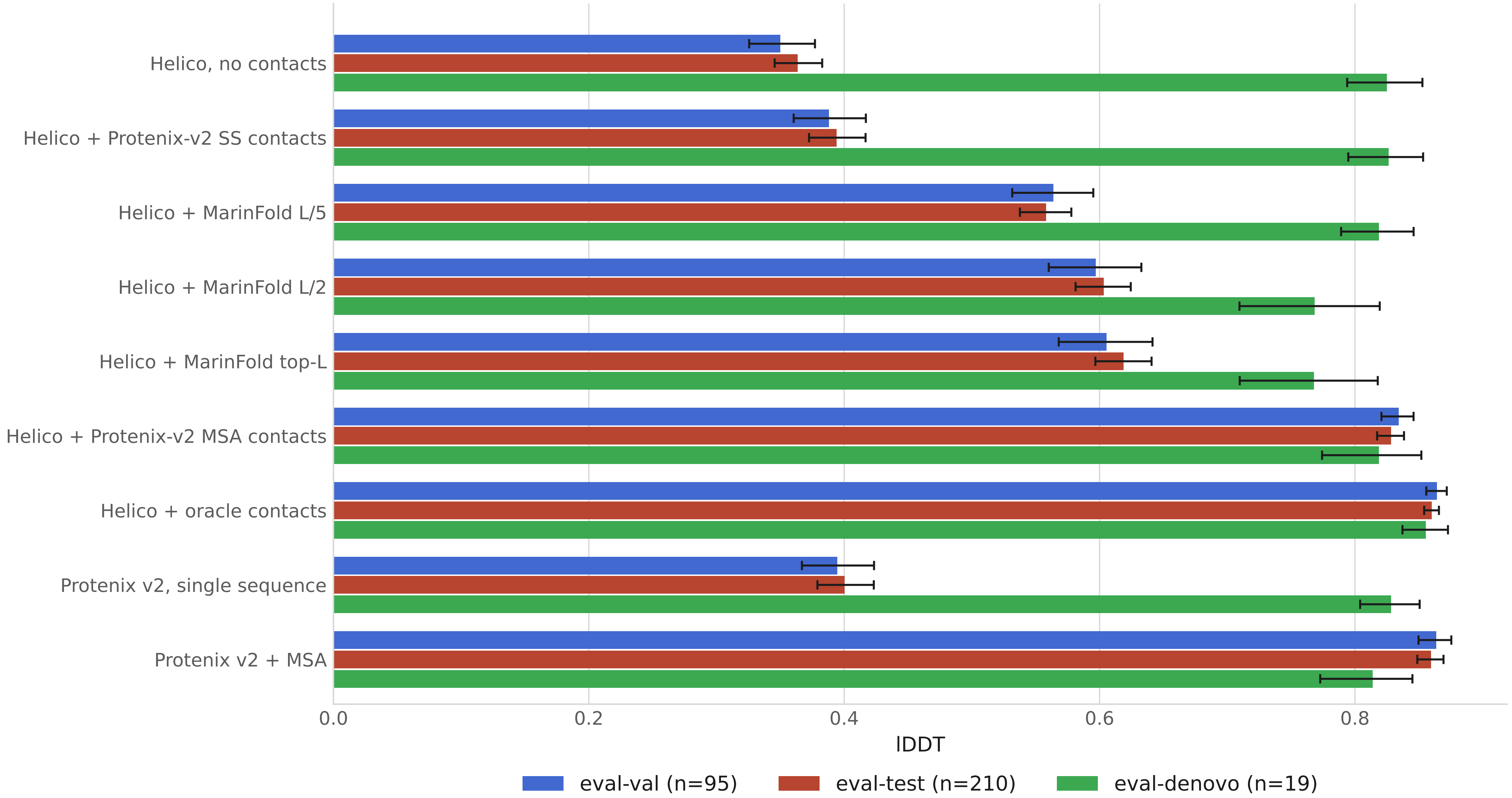
— Contacts from exp232 m2-p06, decontaminated against every protein scored here

— Helico is MSA-free throughout: one sequence and a contact map

— Means carry 95% percentile bootstrap intervals, 10,000 resamples of the proteins

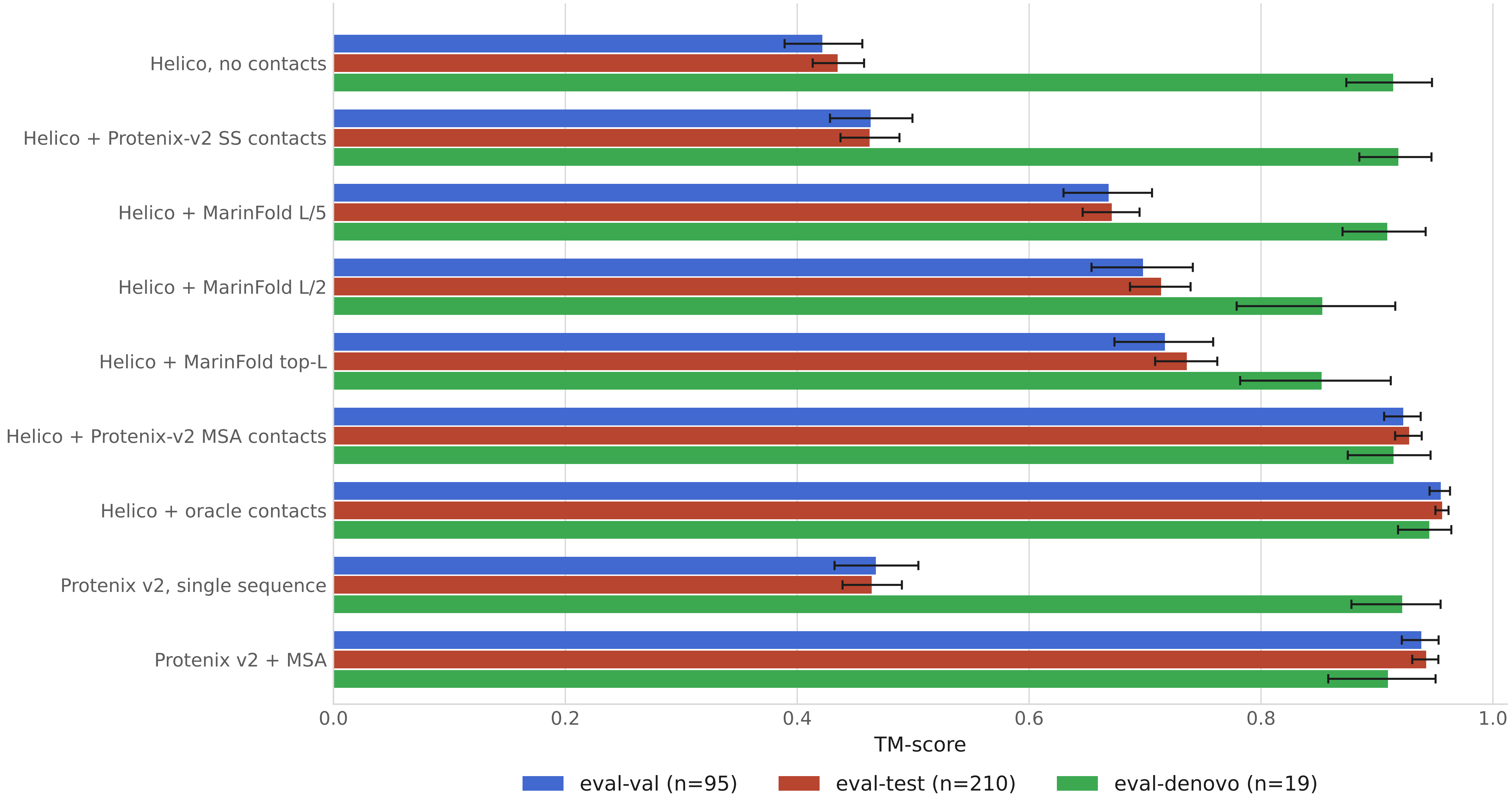
lDDT by conditioning arm

Mean per eval set, higher is better. Error bars are 95% percentile bootstrap intervals over 10,000 resamples of the proteins; every arm sees the same resamples.



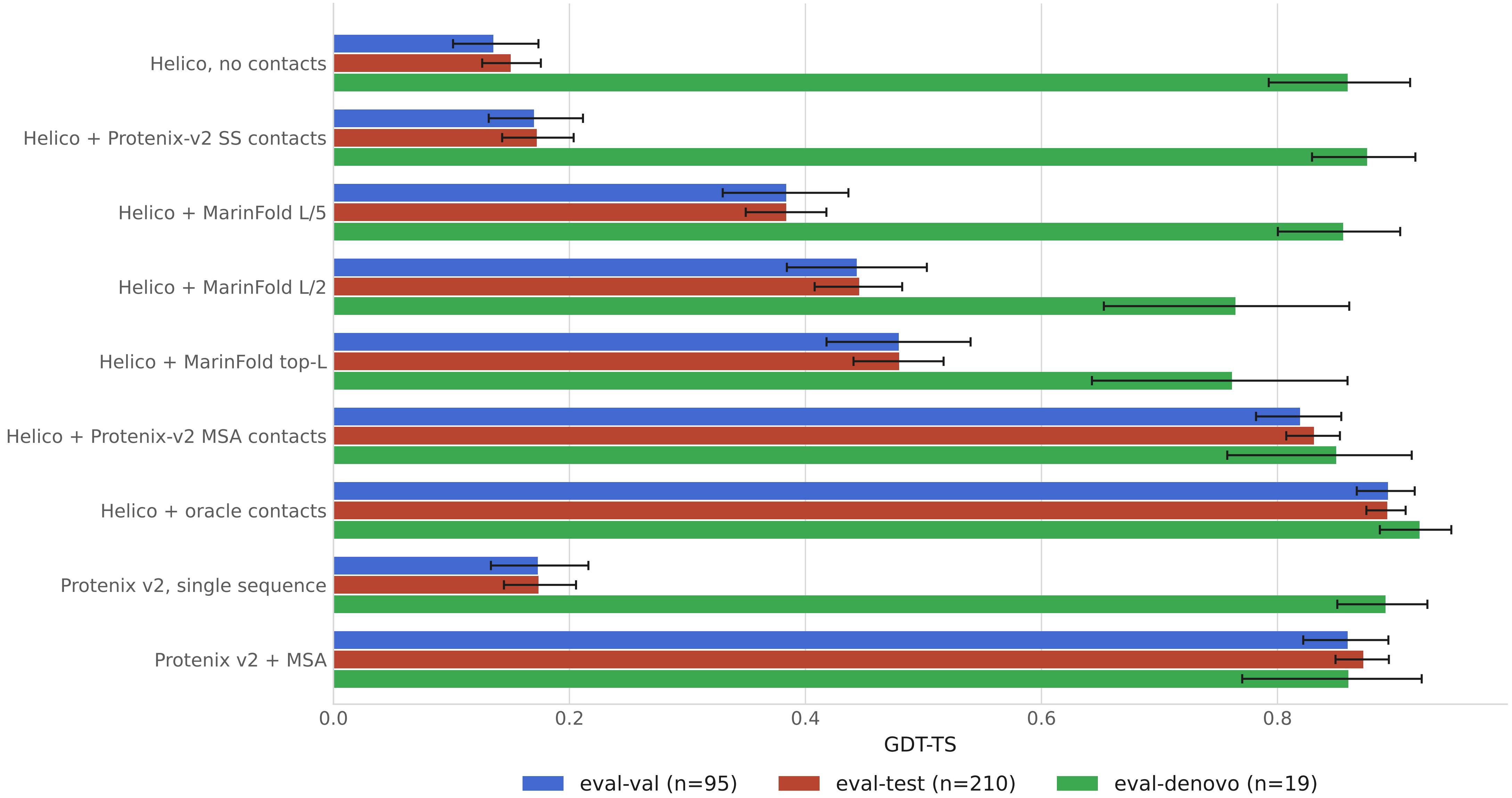
TM-score by conditioning arm

Mean per eval set, higher is better. Error bars are 95% percentile bootstrap intervals over 10,000 resamples of the proteins; every arm sees the same resamples.



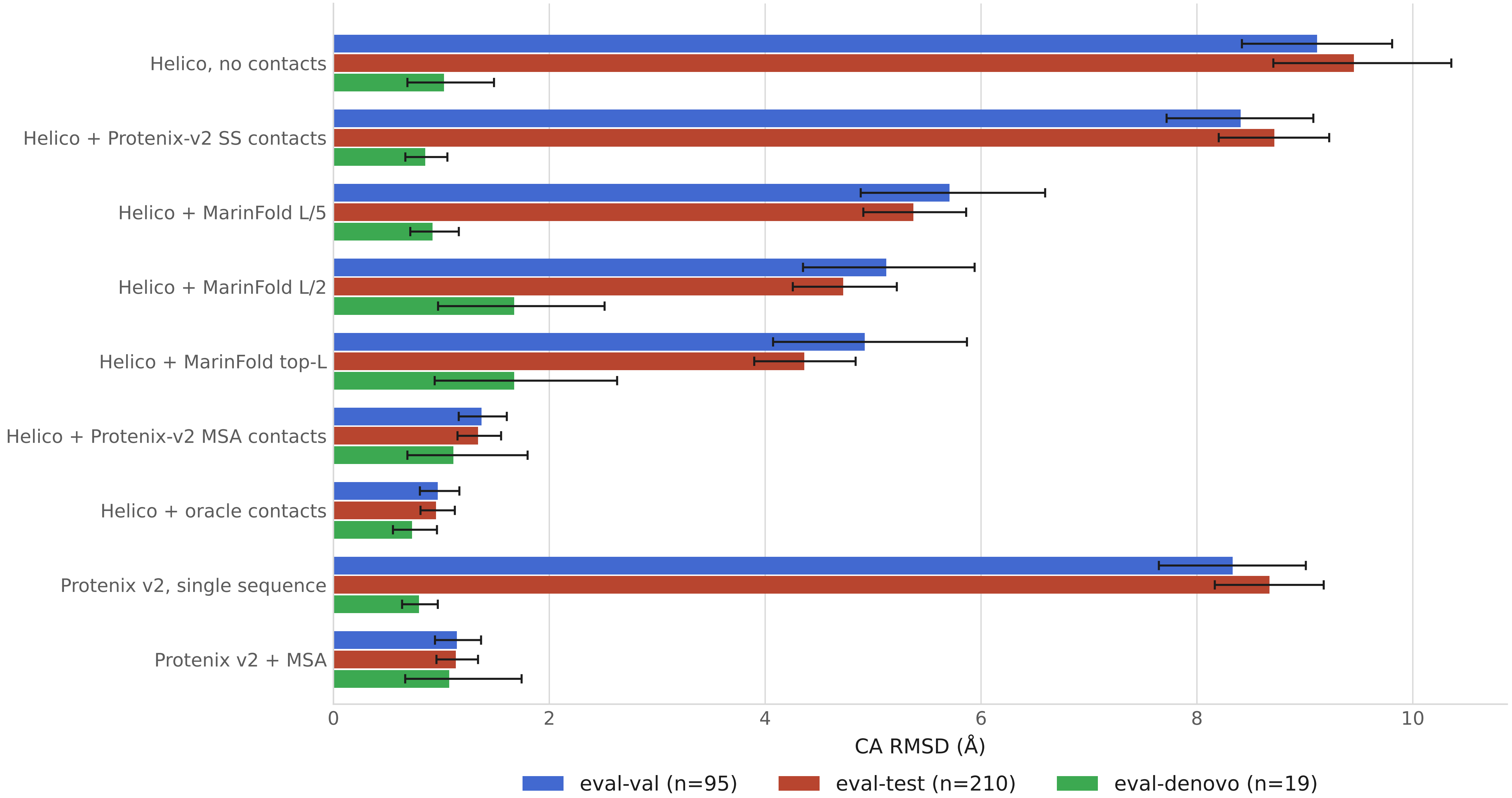
GDT-TS by conditioning arm

Mean per eval set, higher is better. Error bars are 95% percentile bootstrap intervals over 10,000 resamples of the proteins; every arm sees the same resamples.



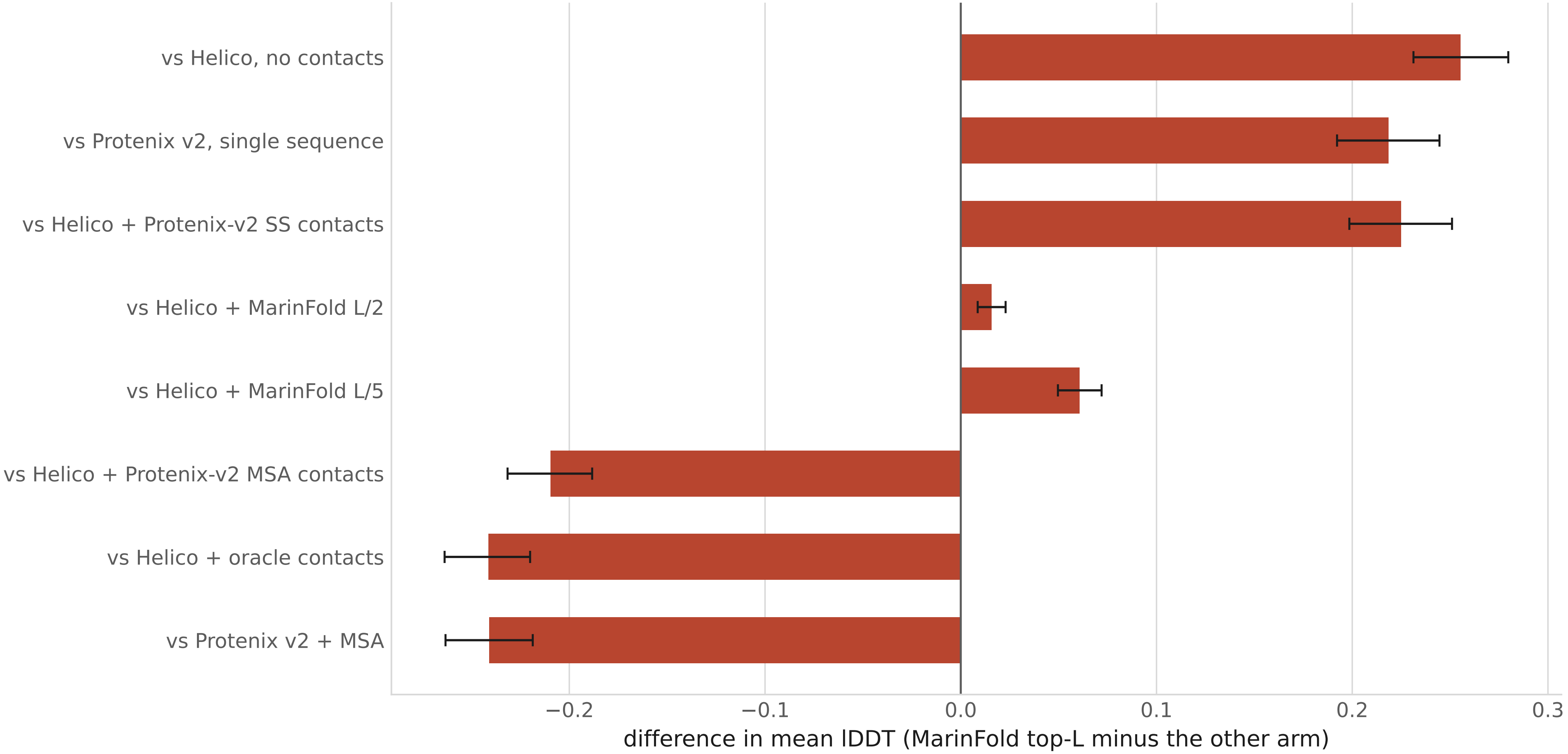
CA RMSD (Å) by conditioning arm

Mean per eval set, lower is better. Error bars are 95% percentile bootstrap intervals over 10,000 resamples of the proteins; every arm sees the same resamples.



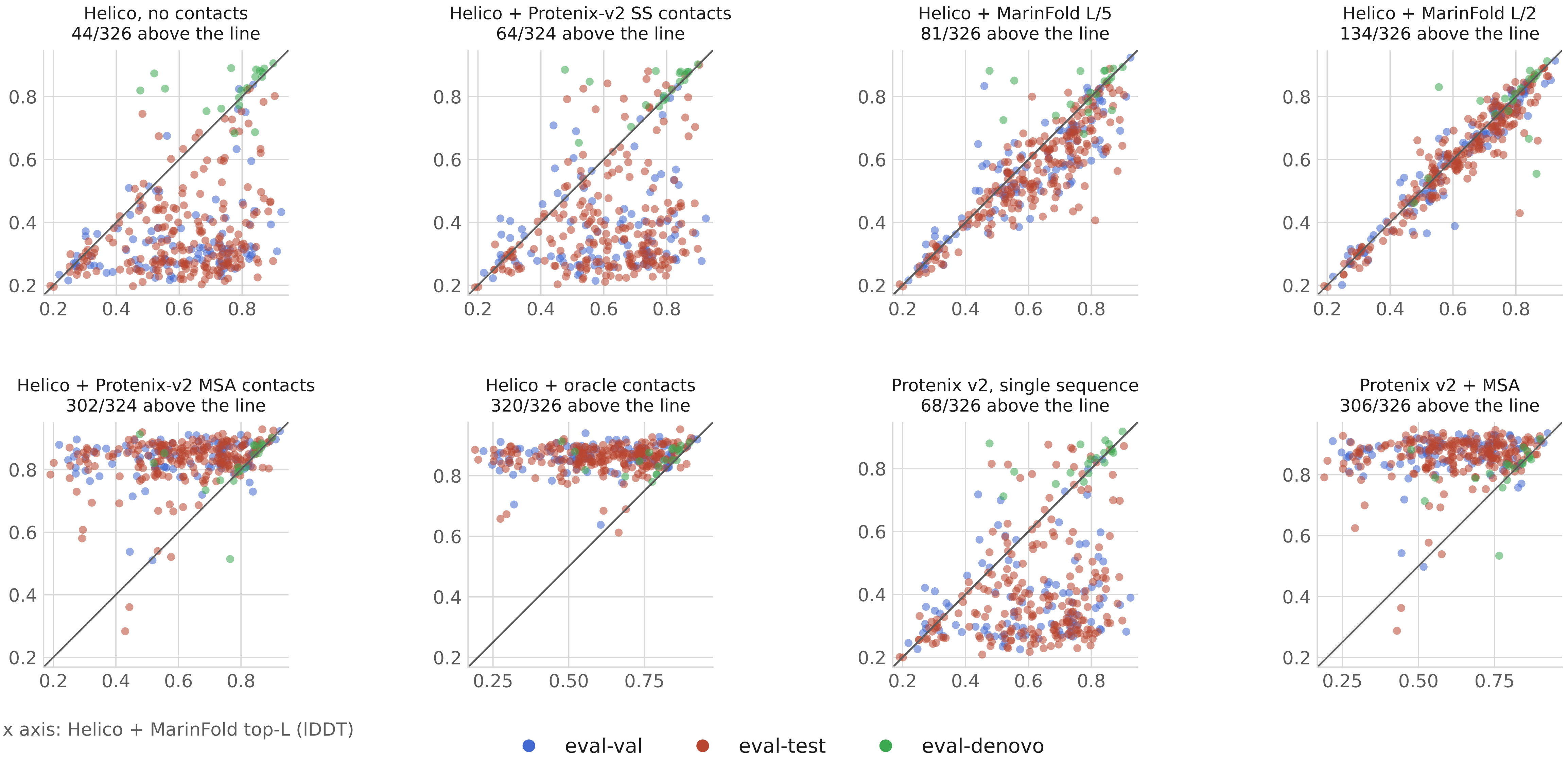
Paired differences against MarinFold contacts

eval-test only. Intervals are 95% percentile bootstrap on the per-target difference, which is narrower than differencing two per-arm intervals.



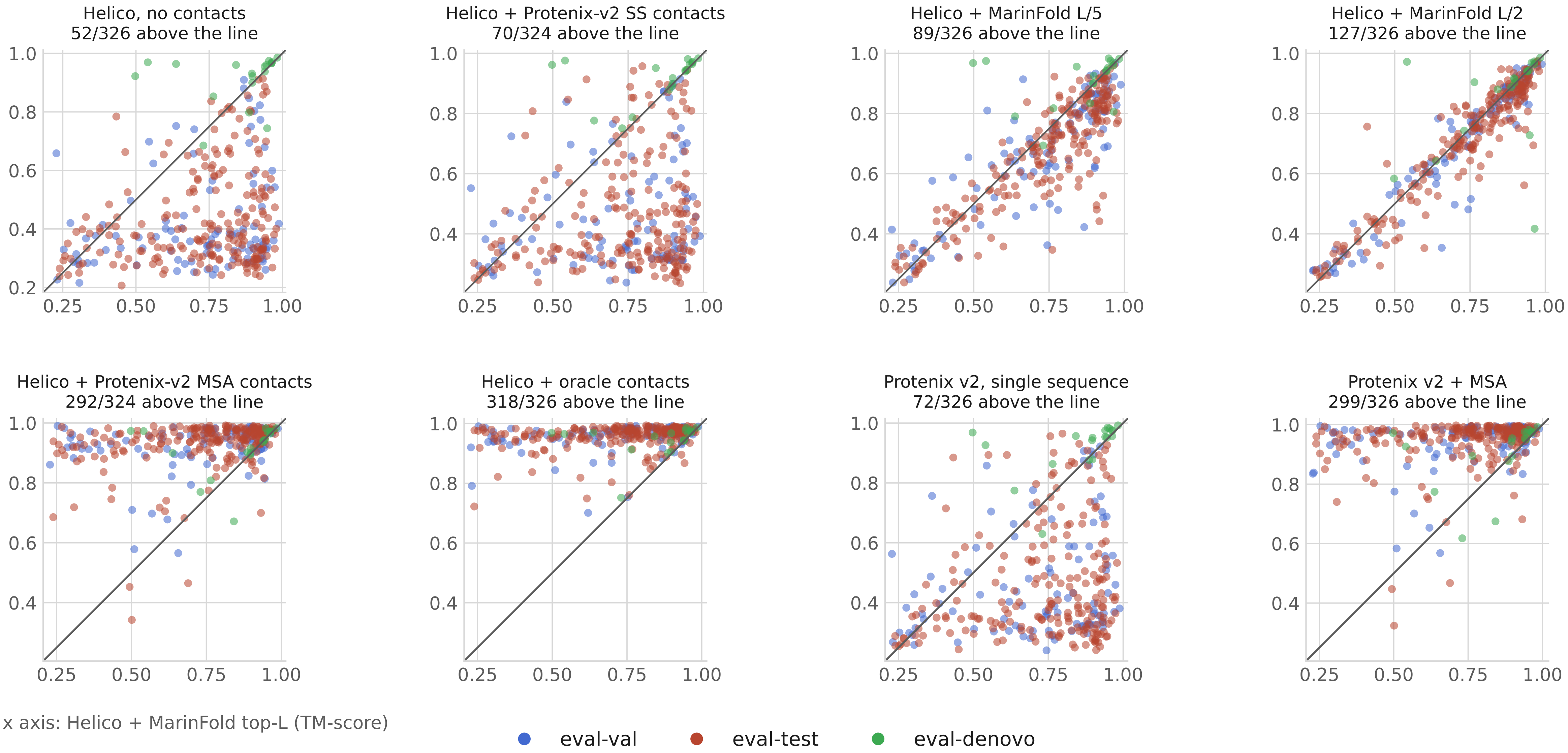
Per-protein lDDT: MarinFold vs each predictor

One point per protein. x is Helico + MarinFold top-L, y is the other predictor; the line is y = x. No bootstrap here — these are individual proteins, not means.



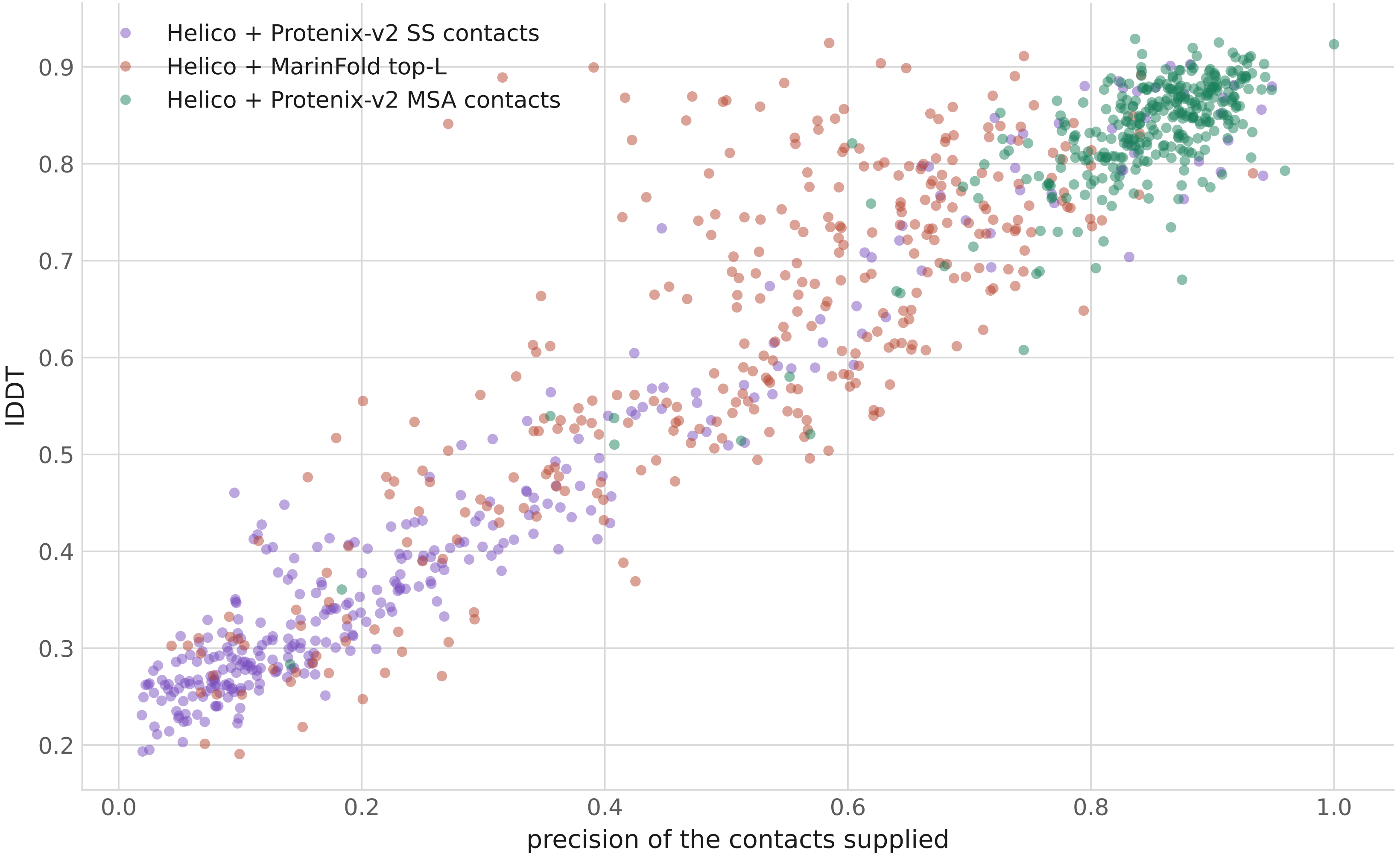
Per-protein TM-score: MarinFold vs each predictor

One point per protein. x is Helico + MarinFold top-L, y is the other predictor; the line is y = x. No bootstrap here — these are individual proteins, not means.



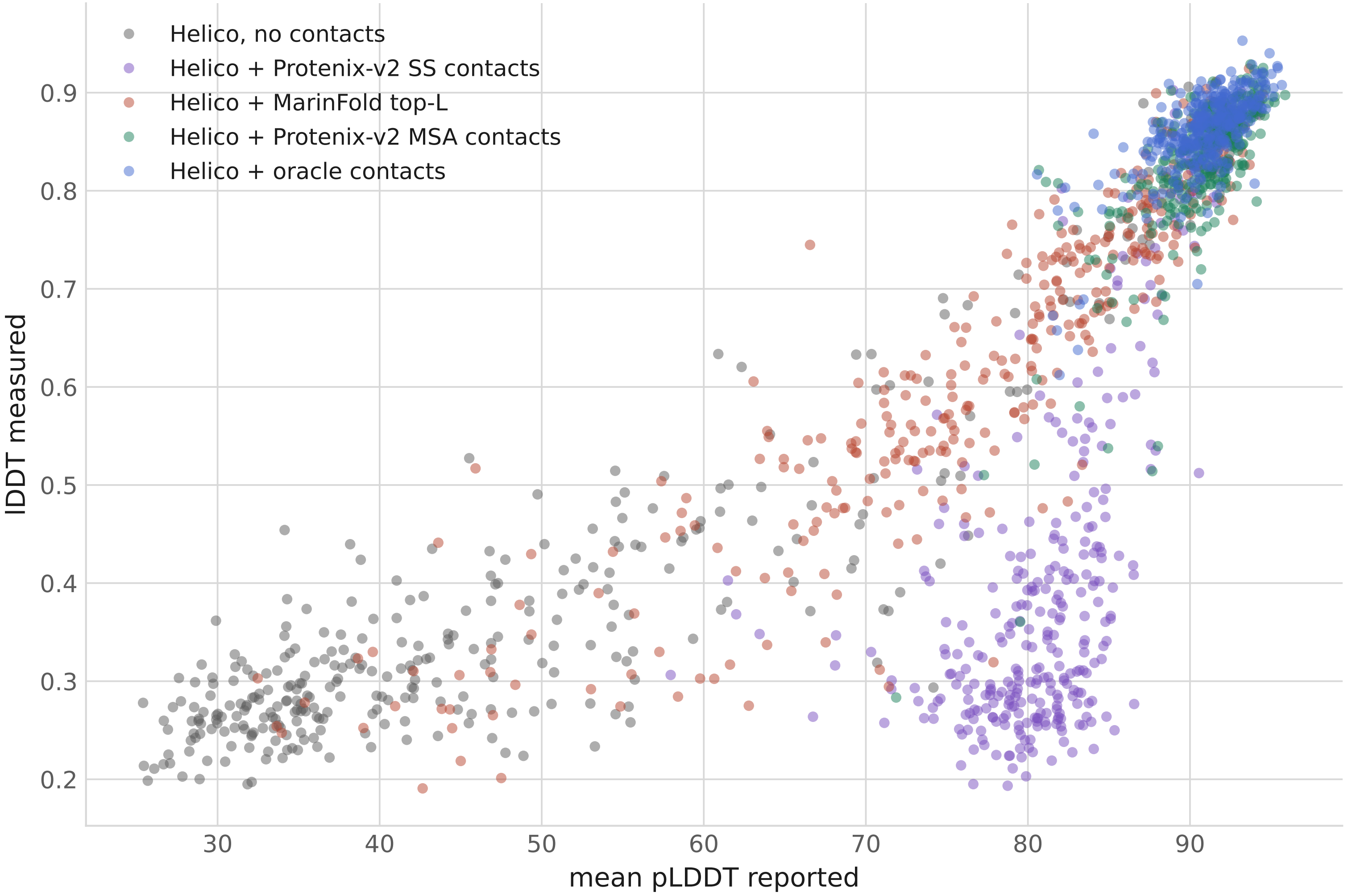
Accuracy against the precision of the contacts supplied

One point per protein, for the three predicted-contact arms. No bootstrap here — these are individual proteins.



Predicted confidence against measured accuracy

One point per protein: the mean pLDDT Helico's confidence head reported, against the lDDT that was measured. This is the score used to pick one of the three diffusion samples. No bootstrap here — these are individual proteins.



mean pLDDT · mean lDDT · Spearman

Helico, no contacts
48.7 · 0.386 · 0.81

Helico + Protenix-v2 SS contacts
81.4 · 0.417 · 0.58

Helico + MarinFold top-L
76.5 · 0.623 · 0.93

Helico + Protenix-v2 MSA contacts
90.7 · 0.829 · 0.70

Helico + oracle contacts
90.9 · 0.860 · 0.65

How each number was produced

Helico

contacts-msafree-01 step 6000 · 6 trunk recycles · 1 trunk run · 3 diffusion samples · seed 42 · bfloat16 · no MSA · best of the 3 by the confidence head's own ranking score (pTM for a single chain)

Protenix v2

protenix 2.0.0, model protenix-v2 · 10 trunk recycles · 1 seed · 5 diffusion samples · 200 diffusion steps · top-1 of the 5 by the model's own ranking score. More inference compute than the Helico arms get, which is conservative here.

MarinFold contacts

exp232 m2-p06 step 145199 · 100 rollouts per protein · occurrence-frequency voting · cut at top-L, L/2, L/5 of the prompt length

Hardware

one NVIDIA H100 80GB per worker, 8 workers · ~13 s per protein per Helico arm

Metrics

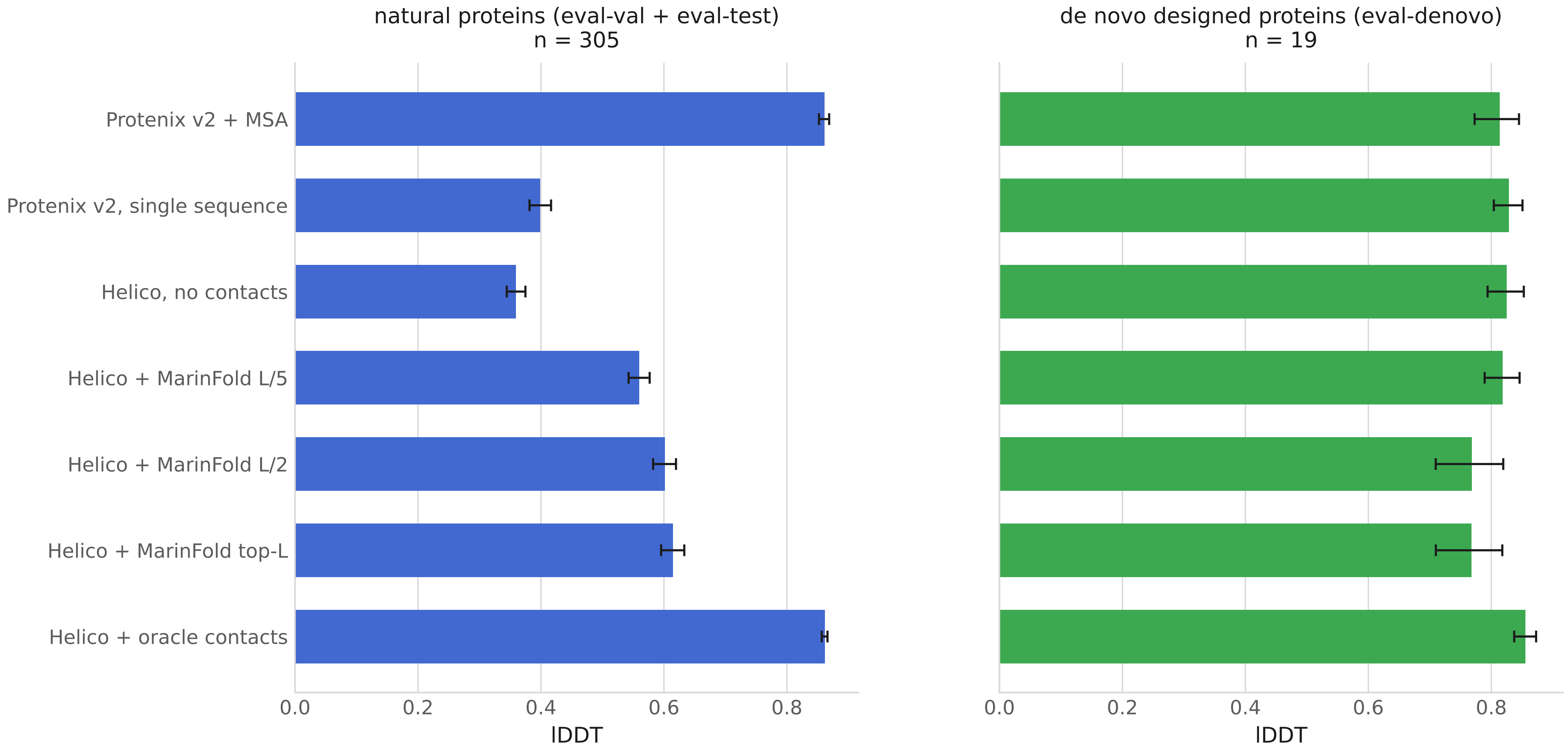
lDDT over all matched atoms; TM-score, GDT-TS and RMSD over one atom per residue (CA), so RMSD is a CA RMSD

Excluded

9 of 333 units: 7 whose contact index map could not be verified, 2 whose Protenix prediction covered under 90% of ground-truth atoms

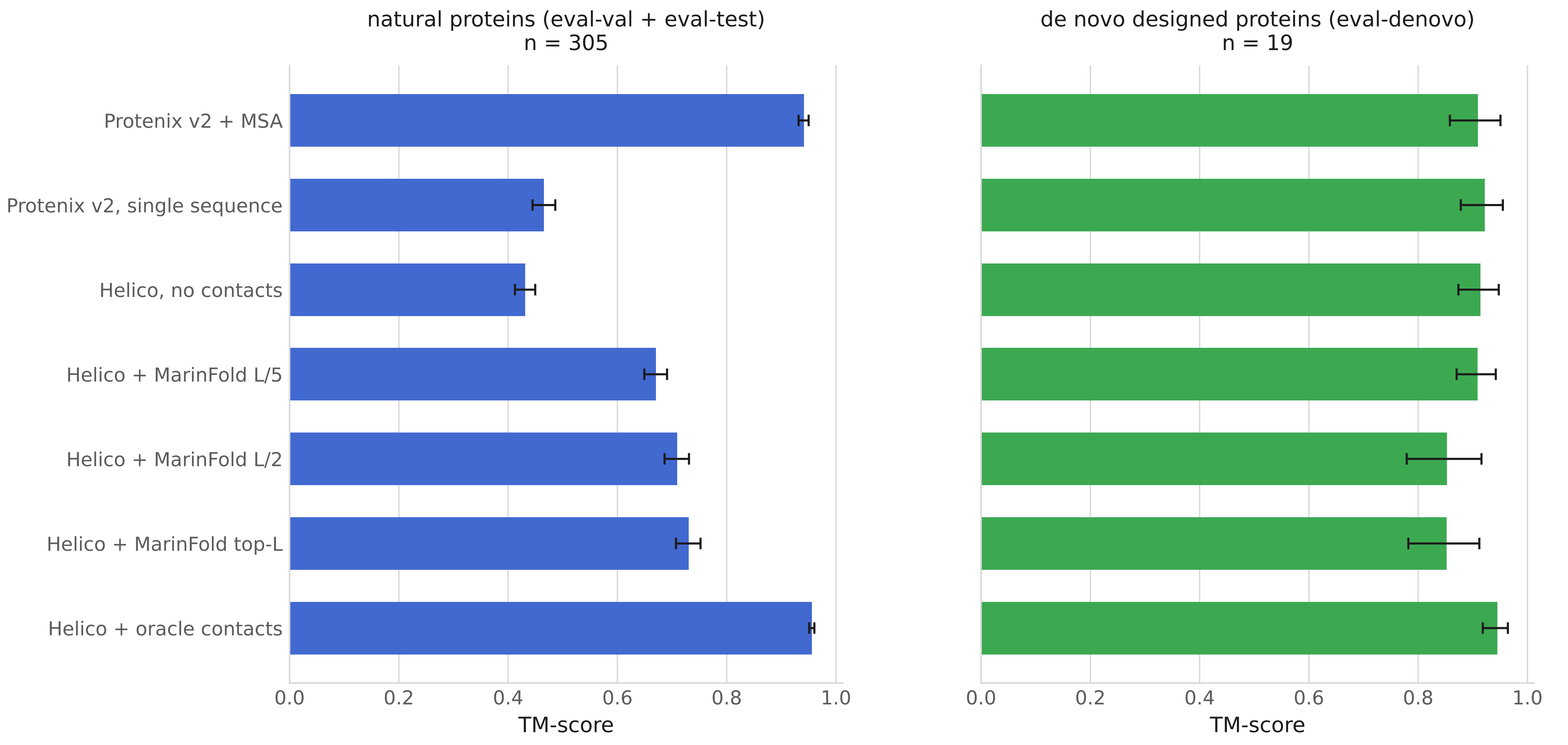
lDDT: natural against designed proteins

Mean per group, higher is better. Error bars are 95% percentile bootstrap intervals over 10,000 resamples of the proteins; every arm sees the same resamples. Both panels share an x range.



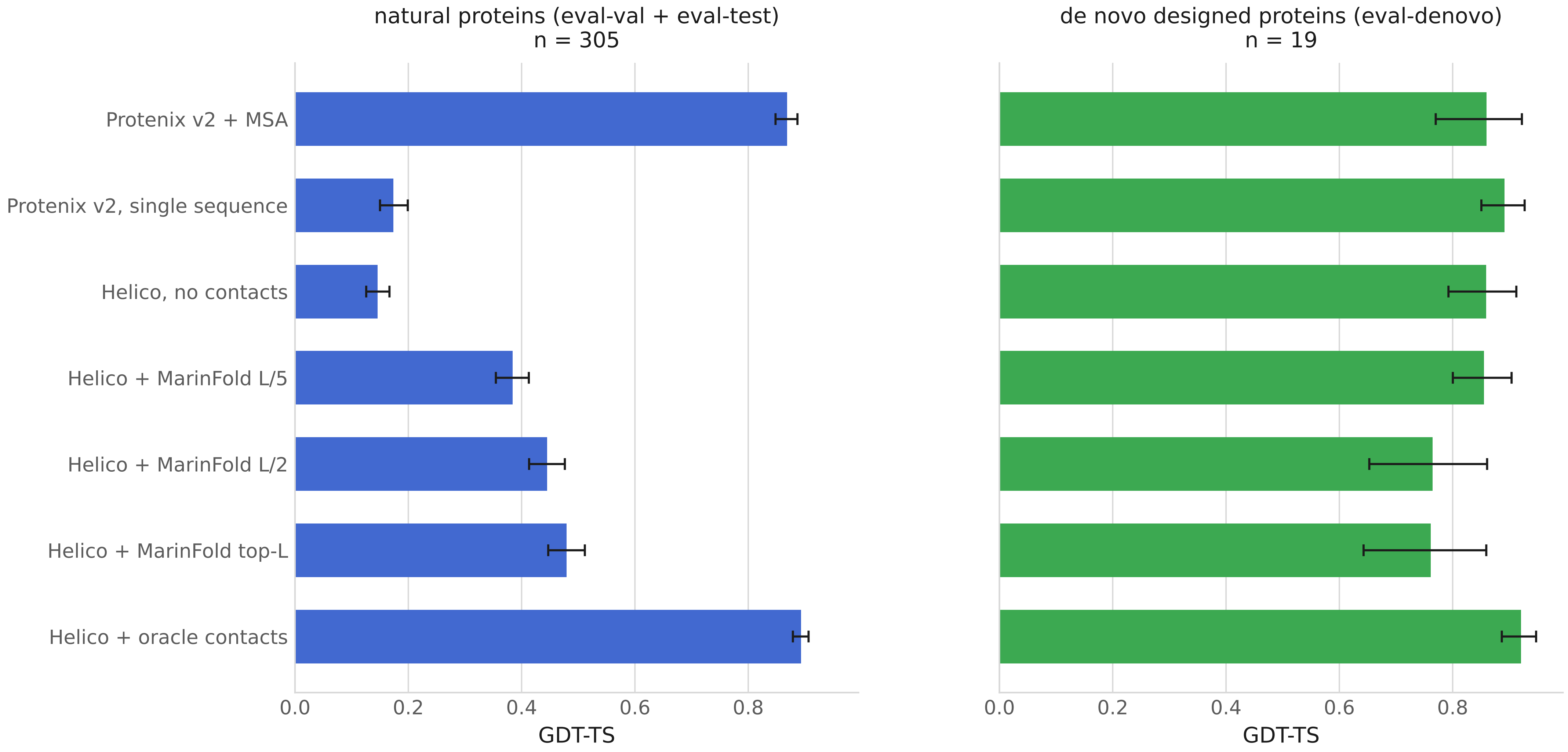
TM-score: natural against designed proteins

Mean per group, higher is better. Error bars are 95% percentile bootstrap intervals over 10,000 resamples of the proteins; every arm sees the same resamples. Both panels share an x range.



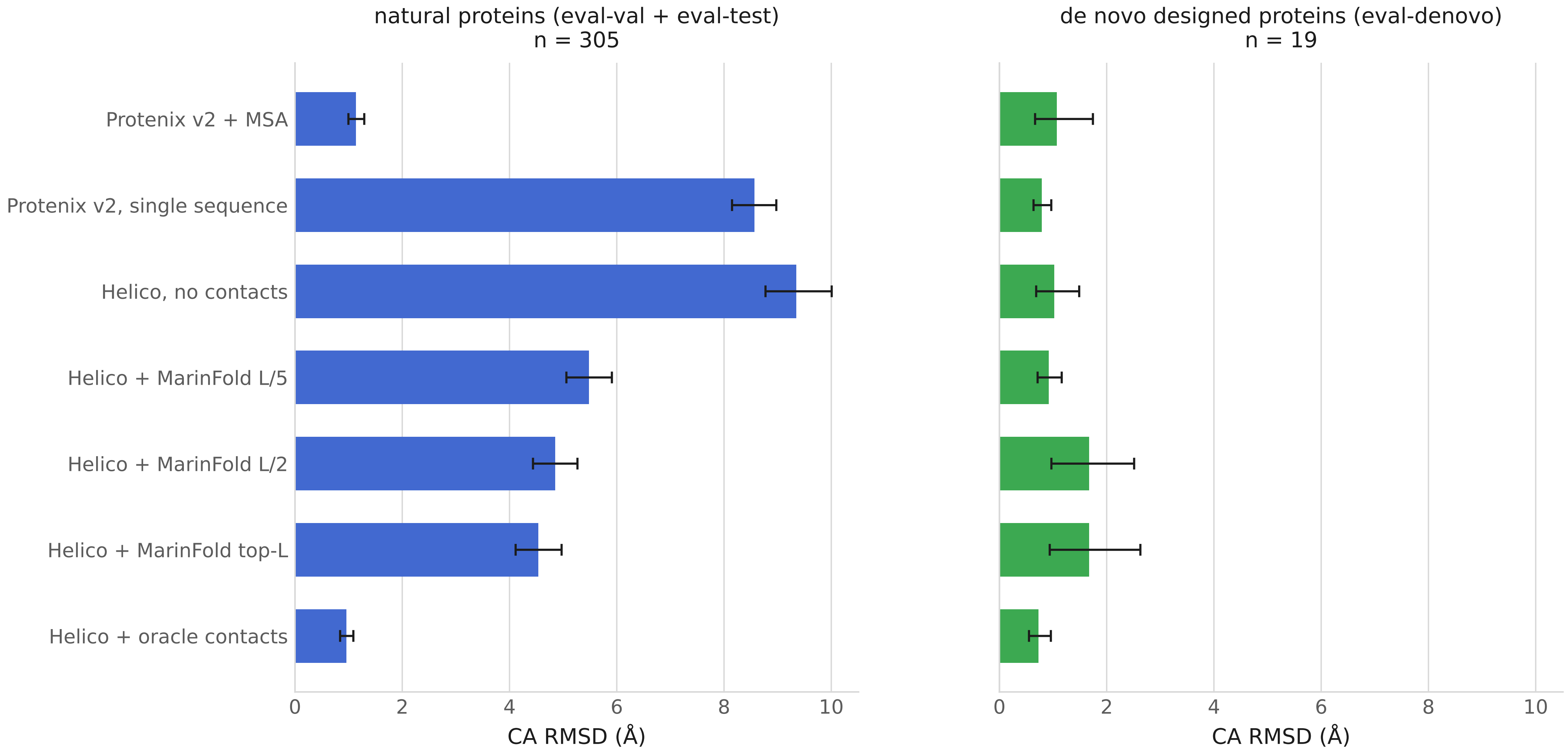
GDT-TS: natural against designed proteins

Mean per group, higher is better. Error bars are 95% percentile bootstrap intervals over 10,000 resamples of the proteins; every arm sees the same resamples. Both panels share an x range.



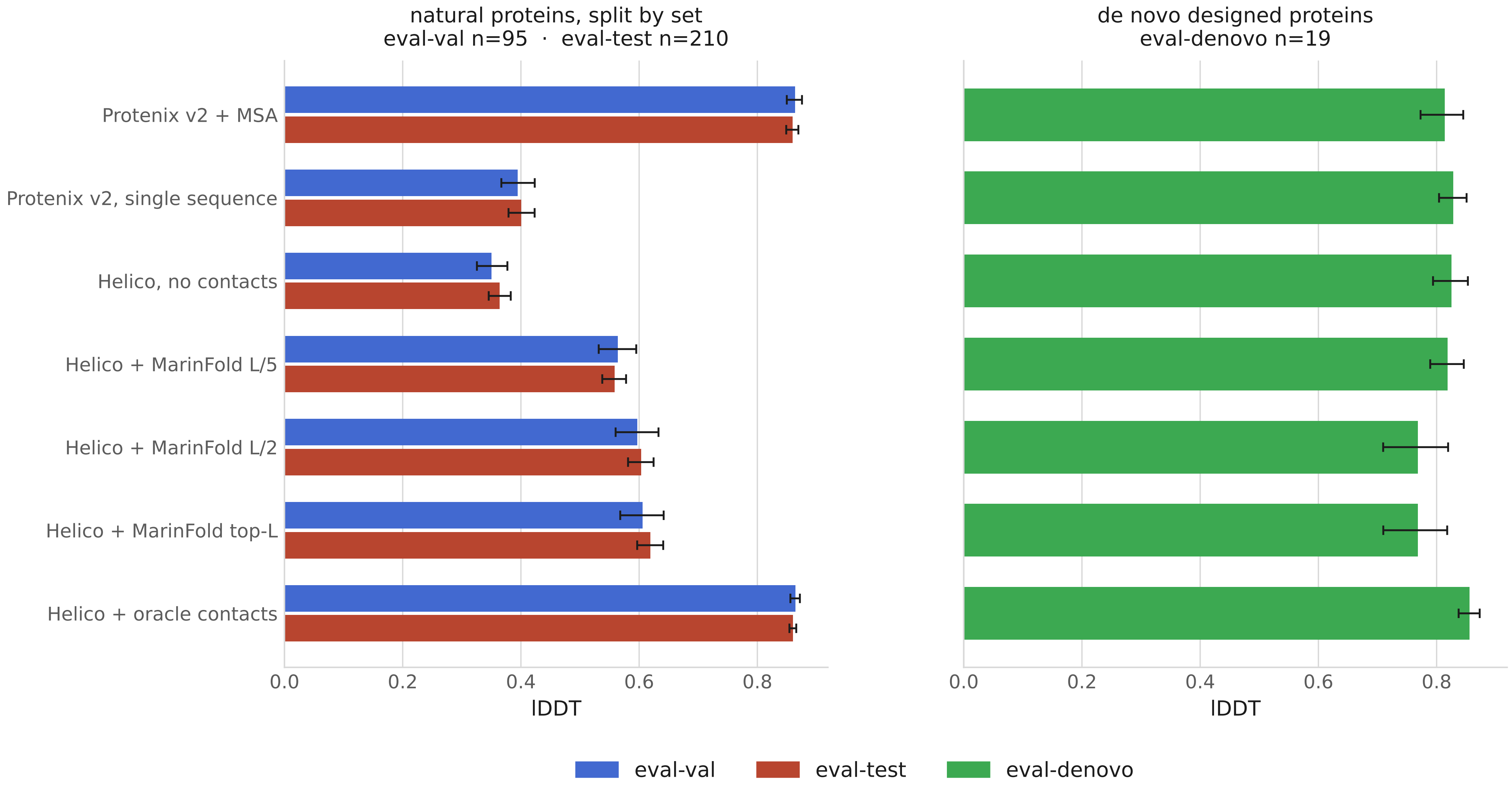
CA RMSD (Å): natural against designed proteins

Mean per group, lower is better. Error bars are 95% percentile bootstrap intervals over 10,000 resamples of the proteins; every arm sees the same resamples. Both panels share an x range.



lDDT: eval-val and eval-test shown separately

Mean per set, higher is better. eval-val is the working set, eval-test the held-out one. Error bars are 95% percentile bootstrap intervals over 10,000 resamples of the proteins; every arm sees the same resamples. Both panels share an x range.



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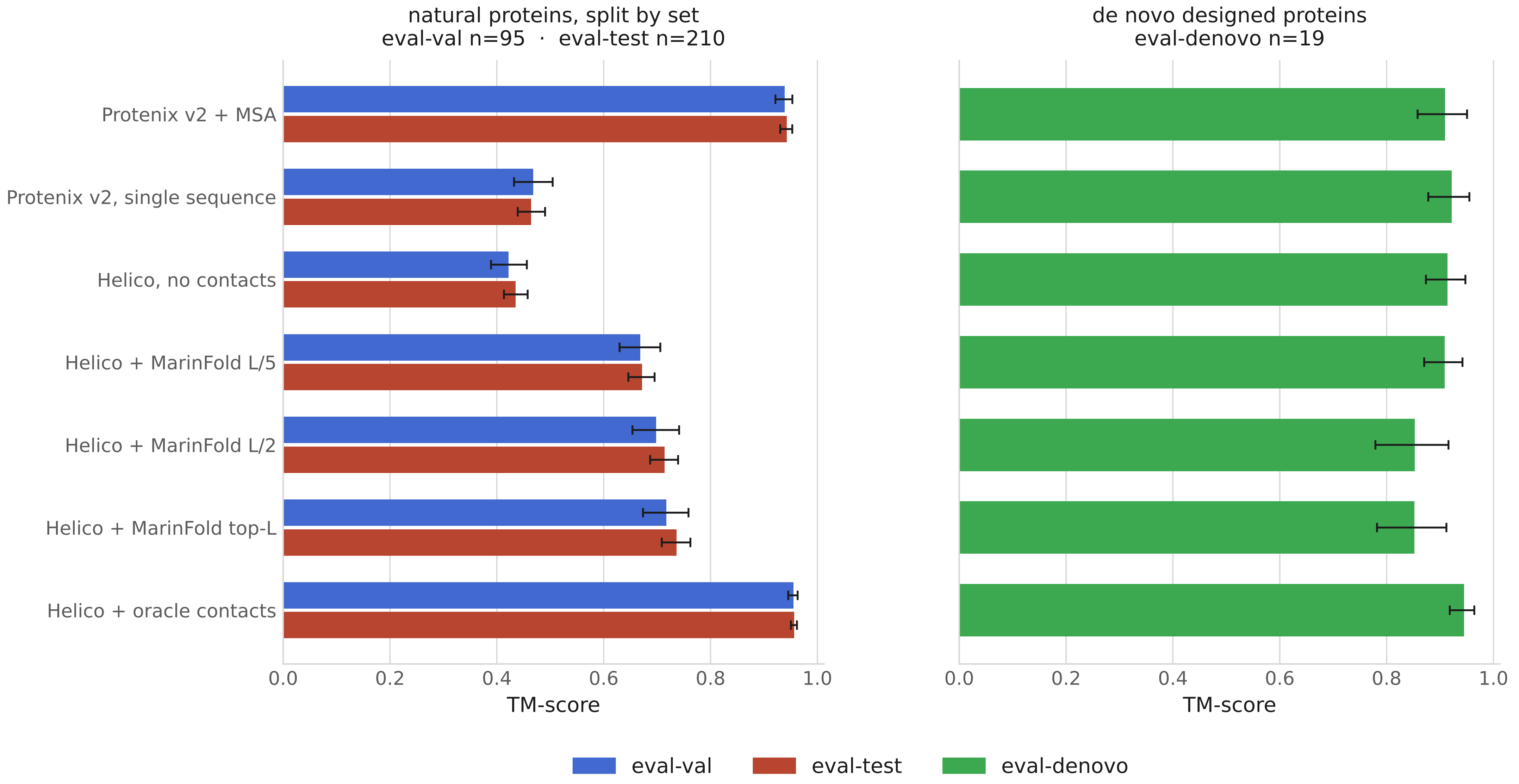

GDT-TS: eval-val and eval-test shown separately

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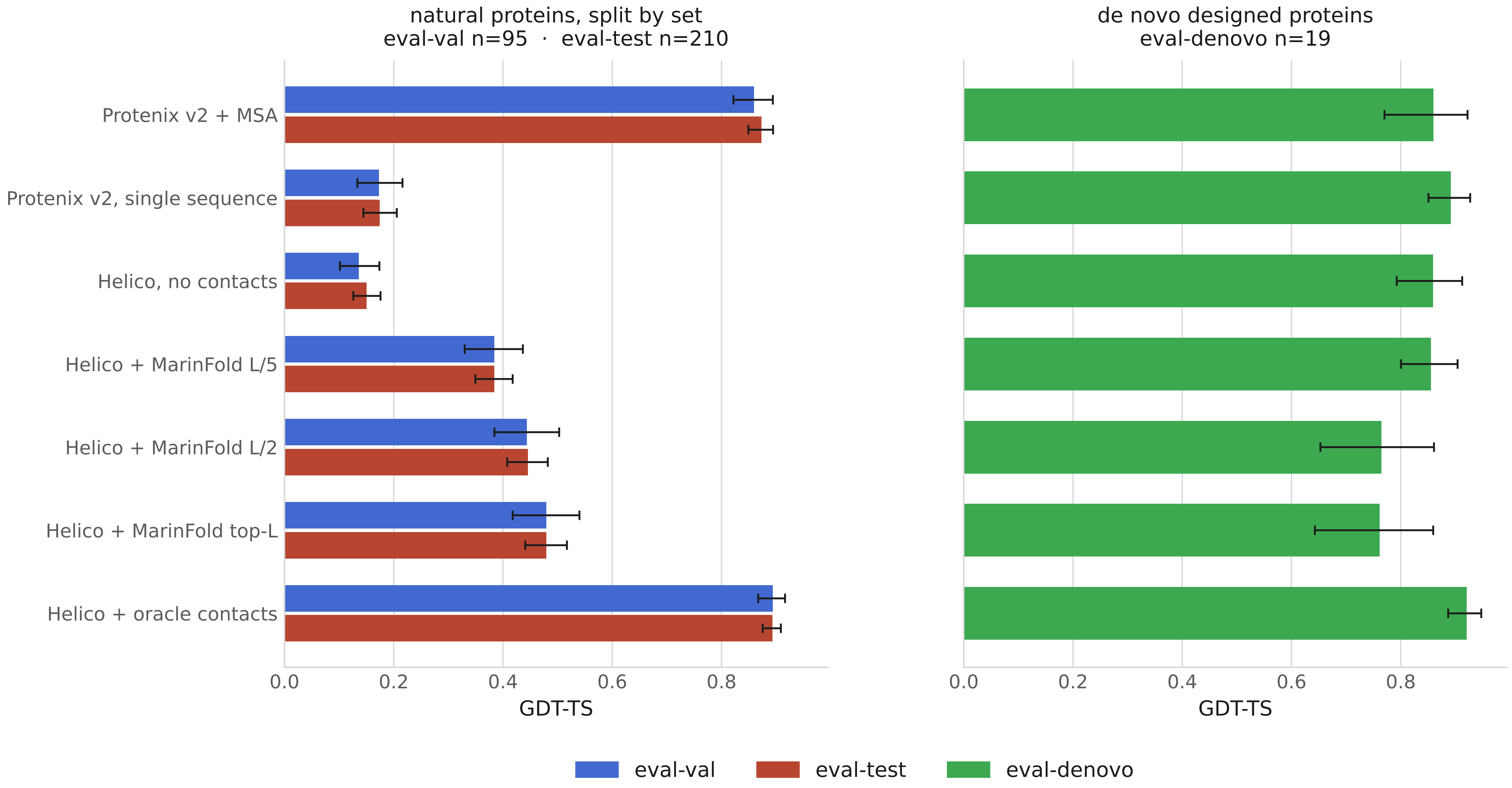

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